



# **Wiz's Journey: Mastering the States of Matter**

## **Producing a Serious Game to Aid in Teaching Matter and Their Phase Changes to Grade 6 Learners**

A Thesis

presented to

the Department of Software Technology

College of Computer Studies

De La Salle University

In partial fulfillment

of the requirements for the degree of

Bachelor of Science in Interactive Entertainment

Major in Game Development

by

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December 7, 2025

## ABSTRACT

Chemistry education contains foundational concepts that are often perceived as difficult for students to understand, particularly in terms of their application to real-world situations. This difficulty may lead to cognitive overload, especially when instruction lacks opportunities for practical engagement. Interactive learning approaches allow students to actively apply and develop their analytical skills while discovering how chemistry concepts relate to everyday experiences.

This study explores the use of Game-Based Learning (GBL) through the development of *Wiz's Journey: Mastering the States of Matter*, an educational serious game designed as a learning material for Grade 6 students under the MATATAG curriculum in the Philippines. The game focuses on physical and chemical changes of matter, expanding students' understanding beyond the three common states of matter. Through interactive gameplay, students engage with concepts such as phase changes and heat transfer while learning to distinguish physical changes from chemical changes.

The primary goal of this study is to improve students' conceptual understanding, motivation, and emotional engagement by reinforcing hands-on learning and real-world applications of chemistry concepts. This project aims to provide both students and teachers with an accessible and engaging instructional resource that supports visualization, experimentation, and the development of analytical skills in basic chemistry education.

**Keywords:** physical and chemical changes, game-based learning, educational serious game

**Sustainable Development Goals:** 4: Quality Education, 11: Sustainable Cities and Communities, 12: Responsible Consumption and Production

## ABSTRAK

Ang pag-aaral ng kimika ay naglalaman ng mga pundamental na konsepto na kadalasang itinuturing na mahirap unawain ng mga mag-aaral, lalo na pagdating sa aplikasyon nito sa totoong buhay. Ang ganitong kahirapan ay maaaring magdulot ng cognitive overload, partikular kung kulang ang mga aralin sa praktikal at interaktibong gawain. Sa pamamagitan ng interaktibong pagkatuto, aktibong nahahasa ng mga mag-aaral ang kanilang kasanayan sa pagsusuri at mas nauunawaan nila kung paano nagagamit ang mga konsepto ng kimika sa pang-araw-araw na buhay.

Sinisiyasat ng pag-aaral na ito ang paggamit ng Game-Based Learning (GBL) sa pamamagitan ng pagbuo ng *Wiz's Journey: Mastering the States of Matter*, isang pang-edukasyong seryosong laro na idinisenyo bilang pantulong na materyal sa pagkatuto para sa mga mag-aaral sa Baitang 6 sa ilalim ng MATATAG kurikulum sa Pilipinas. Nakatuon ang laro sa pisikal at kemikal na pagbabago ng materyal, na nagpapalawak sa kaalaman ng mga mag-aaral lampas sa tatlong karaniwang kaanyuan ng matter. Sa pamamagitan ng interaktibong gameplay, natutuhan ng mga mag-aaral ang mga konsepto tulad ng pagbabago ng estado at paglipat ng init, habang malinaw na natutukoy ang kaibahan ng pisikal at kemikal na pagbabago.

Layunin ng pag-aaral na ito na mapabuti ang pag-unawa, motibasyon, at emosyonal na pakikipag-ugnayan ng mga mag-aaral sa pamamagitan ng hands-on na pagkatuto at aplikasyon ng mga konsepto ng kimika sa mga sitwasyong hango sa totoong buhay. Nilalayon din ng proyektong ito na magsilbing kapaki-pakinabang na materyal para sa mga guro at mag-aaral upang mas mapadali ang pagkatuto ng kimika sa isang biswal, interaktibo, at makabuluhang paraan.

**Mga susing salita:** pisikal at kemikal na pagbabago, game-based learning, pang-edukasyon na laro

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background of the Study

Science education at the elementary level serves as a foundational stage in developing scientific literacy, conceptual reasoning, and problem-solving skills. Within the MATATAG Curriculum of the Department of Education (DepEd), Grade 6 Science introduces learners to fundamental concepts related to materials, including the description of changes of state for solids, liquids, and gases—specifically melting, evaporation, freezing, and condensation—using diagrams and flowcharts. These competencies aim to help learners understand observable phenomena through scientific representations and structured reasoning.

Despite the importance of these competencies, research has consistently shown that students experience difficulty in understanding the particulate nature of matter and the mechanisms underlying phase changes. Learners often memorize definitions of melting or evaporation without fully comprehending the particle-level interactions that explain these transformations. Misconceptions frequently arise regarding particle spacing, motion, and the role of heat transfer in altering states of matter (Adadan, 2013; Paik, Kim, Cho, & Park, 2004). These challenges are compounded by the abstract nature of microscopic representations, which are not directly observable.

In many classroom settings, demonstrations of phase change are limited to water-based experiments, such as melting ice or boiling water. While these activities provide observable macroscopic changes, they often lack explicit integration with particle-level models, resulting in fragmented conceptual understanding. Furthermore, access to laboratory materials for demonstrating processes such as sublimation or deposition is limited at the elementary level, restricting exposure to diverse examples of state change.

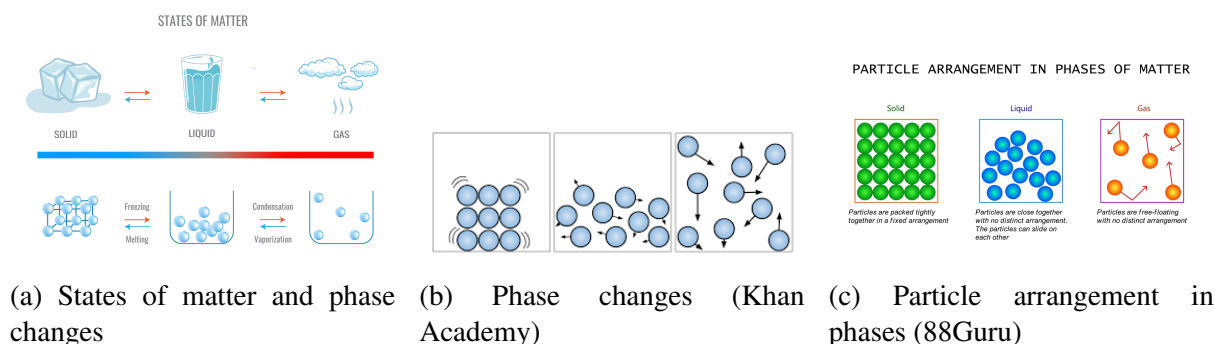


Figure 1.1: Examples of phase change diagrams used in science education

Game-Based Learning (GBL) has emerged as a promising instructional approach that integrates interactive systems with curricular objectives. Educational games can provide dynamic visualizations, immediate feedback, and experiential learning opportunities that reduce cognitive overload and promote conceptual change (Plass, Homer, & Kinzer, 2015). When designed with explicit learning outcomes and aligned assessments, digital games may enhance engagement and deepen understanding, particularly in abstract scientific domains.

This study introduces *Wiz’s Journey: Mastering the States of Matter*, a 2D educational platformer developed as a supplementary and enrichment learning tool aligned with the MATATAG Grade 6 Science competency on changes of state. The game integrates explicit terminology (e.g., “Solid → Liquid: Melting”) while allowing learners to discover appropriate phase changes through environmental problem-solving. Central to its design is an interactive particle-lattice minigame—a core mechanic of the gameplay—in which learners manipulate bonds and particle spacing to simulate state transitions. By directly engaging with representations similar to those used in classroom diagrams, learners repeatedly encounter visual models of particle behavior in a contextualized, goal-driven environment.

Given the increasing emphasis on learner-centered instruction under the MATATAG framework, there is a need to explore innovative digital tools that complement formal instruction without replacing it. This study examines whether a platformer-based educational game can support conceptual understanding, engagement, and reinforcement of curriculum-aligned competencies related to phase changes of matter.

## **1.2 Objectives**

### **1.2.1 General Objective**

This study aims to design, develop, and evaluate an educational platformer game that rapidly operationalizes a selected learning competency from the first quarter of the Grade 6 MATATAG Science curriculum. The game will be developed with established pedagogical and assessment strategies in mind to ensure alignment with curriculum standards and learning outcomes. Learner performance and engagement will be evaluated through pretest and posttest analyses to determine the effectiveness of the intervention.

### **1.2.2 Specific Objectives**

Specifically, the study aims to:

1. Design learning outcome mechanics (LOMs) that strategically target both lower-order and higher-order cognitive learning outcomes aligned with the MATATAG Grade 6 Science competency on changes of state.
2. Develop a functional 2D educational platformer game that incorporates the designed LOMs and integrates interactive particle-level simulations to visually represent state transitions.
3. Evaluate the game's efficacy in teaching the selected learning outcomes to Grade 6 learners through structured pre- and post-assessment instruments, engagement surveys, and qualitative feedback from students regarding usability and perceived learning value.

### **1.3 Scope and Limitations of the Study**

This study focuses exclusively on the development and evaluation of *Wiz's Journey: Mastering the States of Matter*, a 2D educational platformer designed to reinforce the Grade 6 Science competency from the MATATAG Curriculum Guide (CG) 2023 for Grades 3-10. Specifically, the game targets the Quarter 1: Materials learning competency that requires students to “describe changes of state for solids, liquids, and gases as melting, evaporation, freezing, and condensation using diagrams and flowcharts.” The game addresses physical changes only and does not include instruction on chemical reactions or other chemical changes.

The game mechanics are designed to reflect the chosen learning outcomes, focusing on interactive methods for handling game content that directly map to the MATATAG competency. During the development phase, which commenced in December 2025 and extended through March 2026, the development team reviewed existing educational materials and pedagogical approaches to identify optimal mechanics and instructional strategies that could be contextualized for Grade 6 students' current knowledge of phase changes.

The game was developed using Unity as the game engine, chosen for its cross-platform compatibility and the development team's proficiency with the tool. The game is deployed on multiple platforms for greater accessibility: primarily on Android mobile devices via Itch.io, as well as on web platforms via Unity Play. This multi-platform approach ensures that the game can reach students regardless of their available device.

The evaluation methodology consisted of two phases: a formative playtest involving a small sample of students in a controlled environment with qualitative observations and surveys for iterative improvements, followed by a summative playtest with a larger sample that incorporated pretest, gameplay, and posttest assessments to measure learning outcomes.

However, recruitment and participation differed from the initially planned scope. The study originally planned to recruit participants from nearby schools including Inchican Elementary School, Seven Pillars Catholic School, and Princeton Academy. In actuality, only Inchican Elementary School was able to provide informed consent and participate in the study. The school provided 5 consenting students: 3 participants for the formative playtest and 18 participants for the summative playtest (compared to the initially promised 30 students). The other targeted schools were unable to participate due to logistical constraints, including lack of response to communication attempts and the timing of the study during the end of the academic school year when students were occupied with moving-up ceremonies from Grade 6 to first-year high school.

The study does not measure long-term retention of learning outcomes or academic performance beyond the duration of the intervention. Additionally, the research evaluates the game as a supplementary enrichment tool and does not employ a control group design, limiting the ability to make direct comparisons with traditional classroom instruction. The study adopts a single-group quasi-experimental design with pre- and post-assessments, engagement surveys, and qualitative interviews focused on student perception and usability feedback.

#### **1.4 Significance of the Study**

This study contributes to elementary science education in the Philippines by examining the integration of interactive digital modeling within a platformer game environment. For learners, *Wiz's Journey: Mastering the States of Matter* provides repeated exposure to particle-level representations within meaningful problem-solving contexts, strengthening conceptual connections between diagrams and observable phenomena related to phase changes. By aligning the game with the MATATAG Grade 6 Science competency, the study demonstrates how educational games can reinforce curriculum-aligned learning objectives in accessible, engaging ways.

For educators and school administrators, this study offers insights into how educational games may complement MATATAG-aligned instruction without replacing formal teaching. The findings may inform future curriculum integration strategies, digital resource development, and professional development initiatives for teachers adopting game-based learning approaches. Given the resource constraints faced by many Philippine schools, this game's availability on multiple platforms (web and mobile) addresses accessibility barriers and demonstrates feasibility of deploying educational technology in resource-limited contexts.

For researchers and the broader educational community, this study adds to the growing body of literature on game-based learning in science education, particularly within the

Philippine educational context and addressing the K-12 curriculum transition. By examining a discovery-based yet explicitly labeled game design combined with systematic evaluation using pre-post assessments and engagement surveys, the study contributes to ongoing discussions on balancing constructivist learning principles with guided instruction in digital environments. The qualitative feedback from student participants and actual implementation challenges encountered (such as participant recruitment constraints) provide realistic insights into deploying educational games in authentic school settings.

## **CHAPTER 2**

### **REVIEW OF RELATED LITERATURE AND STUDIES**

#### **2.1 Philippine Education Curriculum**

##### **2.1.1 Current Curriculum of Philippines Education**

The Philippines' education system has undergone significant institutional changes in recent years, with the K-12 program serving as the current standardized framework. The K-12 program was implemented to align the Philippine curriculum with international standards and improve compatibility with global education systems (Abragan, Abarcas, Aquino, & Bagongon, 2022). However, recent critical assessments have revealed substantial gaps between intended learning outcomes and actual student performance in classroom settings. Research indicates that discrepancies emerge from multiple factors, including curriculum design, instructional delivery approaches, and alignment of pedagogical objectives with assessment practices (Descartin et al., 2023).

A significant challenge identified by Kilag et al. (2024) is curriculum overload—the K-12 program's extensive content requirements leave insufficient time and space for meaningful student-centered learning. In response to disruptions caused by the COVID-19 pandemic and the shift to remote instruction, the Department of Education introduced the Most Essential Learning Competencies (MELC) framework (Marasigan & Soliman, 2021). The MELC was designed to distill curricula to their core components, ensuring that the most critical competencies were prioritized during instruction (Zalun, 2023; Gallego et al., 2024). Despite these intentions, implementation has revealed persistent challenges—both new issues emerging from MELC adoption and enduring problems that preceded it (Angay, 2023; Gallego et al., 2024).

##### **2.1.2 Status of the Philippine Education**

The broader educational landscape in the Philippines reflects substantial variation in system readiness and implementation capacity across regions and school types. Urban and metropolitan schools, particularly private institutions, generally demonstrate stronger infrastructure and resource availability, enabling more robust curriculum implementation. In contrast, rural and underserved urban schools face systemic capacity constraints that compromise their ability to deliver the full breadth of competencies outlined in national curriculum frameworks (Kilag et al., 2024).

Teaching force preparedness remains a significant concern. While Filipino teachers demonstrate strong commitment and pedagogical dedication, many lack formal professional development in contemporary instructional methodologies—particularly those integrating technology, differentiated instruction, and competency-based assessment (Abragan, Abarcas, Aquino, & Bagongon, 2022). Furthermore, administrative support systems for curriculum implementation vary widely, with limited monitoring and feedback mechanisms in resource-constrained schools to ensure fidelity to curriculum standards (Gallego et al., 2024).

Student learning outcomes in foundational subjects such as science and mathematics have shown modest improvement trends but remain below regional benchmarks. International assessments and national standardized tests indicate persistent achievement gaps correlating with school resource levels and geographic location. These patterns underscore the need for targeted interventions addressing both systemic capacity and localized resource constraints (Descartin et al., 2023).

### **2.1.3 Challenges of Resource Poor Schools**

Resource scarcity presents a fundamental barrier to equitable curriculum implementation in Philippine schools. Many schools, particularly in rural and urban marginalized communities, contend with insufficient learning materials, overcrowded classrooms, and limited access to contemporary instructional tools (Kilag et al., 2024). Abragan and Abarcas et al. (2022) documented that teachers in resource-poor schools lack adequate training in technology integration and have minimal access to information and communication technology (ICT) equipment—tools that could enhance student engagement and maintain focus during instruction.

A critical gap in science education infrastructure is the scarcity of functional science laboratories in resource-poor schools. Many institutions lack dedicated laboratory facilities, essential equipment (microscopes, measuring apparatus, safety equipment), and consumable materials required for hands-on experimentation in chemistry and physics. This absence forces teachers to rely exclusively on theoretical demonstrations and textbook illustrations, severely limiting students' ability to conduct direct observations and experiential learning in abstract scientific domains such as phase changes and matter properties (Kilag et al., 2024).

Curriculum implementation challenges are further compounded by deficiencies in teacher preparation. Curriculum unpacking—the process by which educators analyze curriculum competencies to derive specific learning objectives, instructional materials, and assessment activities—requires specialized training and pedagogical expertise. Many teachers lack formal guidance in this process, resulting in literal interpretation of competency statements without meaningful contextualization or sequencing (Gallego, Andea, et al., 2024). This insufficient preparation,

combined with limited access to resources, leads to compromised mastery of prerequisite concepts, creating foundational gaps that undermine student performance in higher-order learning objectives (Gabriel, Nepomuceno, Kadusale, Taneo, & Cabello, 2022).

Research consistently demonstrates that resource-rich schools—particularly private institutions with superior facility access and professional development infrastructure—achieve significantly higher academic outcomes than their resource-poor counterparts (Angay, 2023). To address this inequity, scholars have called for expanded provision of learning resources across all school contexts. The present study responds to this imperative by developing an educational platformer game focused on reinforcing Grade 6 competencies related to phase changes and states of matter. By leveraging the wide availability of mobile devices and web-accessible platforms, this game-based intervention can reach students regardless of whether their schools possess traditional science laboratory facilities. The game provides interactive, visual simulations of molecular-level phenomena aligned with MATATAG competencies, enabling students to engage with abstract scientific concepts through accessible gaming technology that requires minimal institutional infrastructure beyond basic internet connectivity.

## **2.2 Chemistry and Science Education**

### **2.2.1 Particle Models and Phase Transitions in Elementary Learning**

Understanding the particulate nature of matter and phase transitions is fundamental to chemistry and science literacy. Samarapungavan, Bryan, Staudt, and colleagues (2023) explored kindergarten and early elementary students' learning of simple particle models to explain the properties and behavior of matter in solid, liquid, and gas states during phase transitions. Their research found that when students are explicitly taught to use particle models as explanatory frameworks, they develop more accurate mental representations of microscopic processes underlying observable macroscopic phenomena. The study emphasizes that science instruction focusing solely on concrete, directly observable phenomena without explicit instruction in particulate models limits students' ability to understand abstract concepts fundamental to chemistry.

#### **2.2.2 Misconceptions in Melting and Freezing Cycles**

A critical challenge in science education is bridging the gap between macroscopic observations and microscopic explanations. Smith and Villarreal (2015) assessed 155 freshman general chemistry students' misconceptions about particle position and arrangement during the melting-freezing cycle using the Melting Cycle Instrument, which employs particulate-level



representations. Their findings revealed that many students maintained inaccurate mental models about how particles behave during phase transitions, particularly regarding particle spacing and motion during state changes. These misconceptions persist even after formal instruction, suggesting the need for innovative pedagogical approaches that make particle-level interactions more salient and comprehensible to learners across educational levels.

### **2.2.3 Remediation of Matter Misconceptions Through Knowledge Revision**

The persistent nature of matter misconceptions has prompted researchers to investigate interventions designed to remediate student errors. Korur, Korumaz, and Avci (2025) developed the Knowledge Revision Components (KReC) framework, which integrates refutation texts and advanced organizer materials to explicitly address and eliminate misconceptions about matter and physical/chemical changes. Their research demonstrated that systematic, theory-based approaches to misconception remediation significantly improve student conceptual understanding when learners are directly confronted with the inadequacy of their existing mental models. This approach aligns with pedagogical principles emphasizing active engagement with conflicting ideas as a mechanism for conceptual change.

### **2.2.4 Middle School Misconceptions About Matter Cycles and Conservation**

At the middle school level, Arslan (2023) employed a word association test to investigate 457 eighth-grade students' cognitive perceptions and misconceptions regarding matter cycles and environmental phenomena. The study revealed widespread misconceptions about evaporation, condensation, and the conservation of matter during state changes, particularly among students in lower-performing schools with limited access to interactive learning materials. These findings underscore how resource constraints in Philippine schools compound the challenge of teaching abstract science concepts and exacerbate existing achievement gaps. The representation and presentation of scientific concepts significantly influence student understanding, particularly when integrating multiple representational modes—text, diagrams, animations, and interactive manipulatives—to support more robust conceptual development.

## **2.3 Game-Based Learning**

### **2.3.1 Developmental Strategies of Edugames**

Constructivist Learning Theory asserts that knowledge is actively constructed through interaction and experience rather than passively received. Game-based environments embody constructivist principles by placing learners in active problem-solving roles. In platform-based

game environments, players experiment, test strategies, and receive immediate feedback from system responses. This aligns with the constructivist view that learning occurs through engagement and reflection upon action (Kolb, 1984; Bruner, 1966).

Outcome-based methodologies provide a structured framework for developing game-based learning environments aligned with explicit educational competencies. Sison and colleagues (2018) presented an outcome-based methodology for developing game-based learning environments, emphasizing the importance of clear learning objectives that guide both game design and assessment strategies. This approach ensures that gameplay mechanics directly support the achievement of specified educational outcomes, creating coherence between instructional goals and interactive gameplay.

In game development research, iterative design is a widely accepted approach for refining gameplay mechanics, usability, and user experience (Gee, 2003). Iterative development involves repeated cycles of testing, evaluation, modification, and regression validation. Campaign testing refers to structured playthrough sessions aimed at identifying usability issues, progression bottlenecks, and gameplay friction points. Engagement evaluation models assess multiple dimensions including focused attention, satisfaction, perceived learning, and intrinsic motivation. User-centered design principles emphasize incorporating player feedback into successive iterations to improve overall game quality (Prensky, 2001). Educational game development should adopt a quasi-experimental assessment approach with pre- and post-gameplay measures to evaluate learning effectiveness, employing both quantitative metrics (test scores, learning gains) and qualitative methods (engagement surveys, observational feedback) to comprehensively evaluate user experience and educational impact.

### **2.3.2 Effectiveness of games in education**

Multimedia Learning Theory suggests that learning is enhanced when verbal and visual information are presented in coordinated and complementary ways (Mayer, 2001, 2009). Effective multimedia design integrates text, visuals, and animation to support meaningful cognitive processing. Research specifically demonstrates that animations with textual labels significantly enhance learning outcomes in scientific domains where direct observation is limited (Mayer & Moreno, 2002). Educational games that integrate explicit textual labeling (e.g., “Solid → Liquid: Melting”) alongside dynamic visualization can support the formation of coherent mental representations of abstract processes.

Game-Based Learning Theory emphasizes the role of challenge, feedback, and goal-oriented interaction in promoting learner engagement (Prensky, 2001). Educational games combine instructional objectives with gameplay mechanics to foster sustained attention and intrinsic motivation. Engagement is a critical factor in educational technology adoption. Games

that balance challenge and usability are more likely to maintain player interest and promote exploratory interaction (Gee, 2003). Assessment of engagement using structured instruments can determine whether a game builds demonstrates positive user experience.

Experiential Learning Theory posits that learning occurs through cycles of experience, reflection, conceptualization, and experimentation (Kolb, 1984). Simulated environments allow learners to experience phenomena that may not be directly observable in traditional classrooms. By simulating molecular interactions during freezing, melting, and evaporation, educational games can provide experiential exposure to abstract scientific processes aligned with curriculum standards. Outcome-Based Teaching and Learning emphasizes alignment between instructional activities and clearly defined competencies. Game level design can encourage players to apply conceptual knowledge to solve in-game challenges, aligning with higher-order cognitive objectives in frameworks such as Bloom's Taxonomy (Anderson & Krathwohl, 2001).

### **2.3.3 Game-Based Learning in the Philippines**

International research has demonstrated the effectiveness of educational games in enhancing engagement and conceptual understanding in science subjects. Studies evaluating interactive simulations and digital learning games report improvements in learner motivation and positive user experience when multimedia elements and active manipulation are integrated. Additionally, research on usability evaluation models such as MEEGA+KIDS highlights the importance of assessing engagement, focused attention, satisfaction, and perceived learning in educational game contexts (von Wangenheim, Petri, & Borgatto, 2020). Studies on iterative game development emphasize that structured playtesting contributes significantly to improved playability, stability, and user satisfaction (Sison et al., 2018).

Local studies in educational technology within the Philippine context indicate increasing integration of digital learning tools aligned with curriculum standards. Research evaluating instructional games commonly employs usability surveys and engagement instruments to assess learner reception. However, limited research combines structured iterative development practices with formal engagement evaluation within educational game development projects in the Philippine context. This gap supports the relevance of developing and evaluating curriculum-aligned educational games that address the learning needs of Filipino students while demonstrating rigorous development and evaluation methodologies.

## **2.4 Theoretical Framework**

This study is grounded in a hierarchical integration of theoretical frameworks that collectively explain both the pedagogical rationale for game-based learning and the methodological ap-

proach employed in development and evaluation.

### **Tier 1: Primary Theoretical Anchors**

Conceptual Change Theory serves as the primary foundation, particularly given the documented misconceptions about phase changes identified in the research literature (Posner, Strike, Hewson, & Gertzog, 1982; Section 2.2). This theory posits that learners construct initial mental models of phenomena through everyday experiences; these models often conflict with scientifically accurate explanations. Conceptual change occurs not through passive information reception, but through active confrontation with the inadequacy of existing models (Posner et al., 1982). In the context of this study, students hold misconceptions about particle behavior during phase changes; the game is designed to create problem-solving scenarios that expose the limitations of these mental models and provide experiential evidence supporting more accurate particle-level explanations (Samarapungavan et al., 2023).

Constructivism complements Conceptual Change Theory by emphasizing that knowledge is actively constructed through interaction, experience, and reflection rather than passively received (Bruner, 1966; Kolb, 1984). In a constructivist game environment, learners do not receive explicit lectures on phase changes; instead, they manipulate particle spacing and bond strength through the interactive particle-lattice minigame, discovering the relationships between these manipulations and observable state transitions. This active engagement supports the conceptual change process by anchoring abstract ideas (particle motion) to concrete, manipulable experiences within the game (Kolb, 1984).

Multimedia Learning Theory addresses how to represent abstract concepts through coordinated presentation of multiple information channels (Mayer, 2001, 2009). The game integrates text labels (e.g., “Solid to Liquid: Melting”), animated particle representations, and interactive mechanics to convey phase change concepts. Research demonstrates that coordinated text-visual-animation presentations enhance learning outcomes compared to single-modality instruction, particularly for abstract domains where direct observation is limited (Mayer & Moreno, 2002). By simultaneously showing particle lattice structures, temperature effects, and textual terminology, the game leverages multimedia principles to support learner comprehension of microscopic phenomena.

### **Tier 2: Supporting Theories**

Game-Based Learning Theory justifies the use of challenge-based interactive mechanics to foster sustained engagement (Prensky, 2001; Gee, 2003). This theory emphasizes that games

combine immediate feedback, clear goals, balanced challenge-to-skill ratios, and intrinsic motivation mechanisms. In *Wiz's Journey*, these elements manifest as: (1) instant feedback when particles respond to player manipulations, (2) explicit win/fail conditions for phase change challenges, (3) progressive difficulty scaling, and (4) reward systems that reinforce both achievement and conceptual exploration (Prensky, 2001).

Experiential Learning Theory supports the use of simulated environments for learning abstract scientific processes (Kolb, 1984). This theory describes learning as occurring through cyclical experience-reflection-conceptualization-experimentation. The game environment allows learners to experience phase change phenomena that cannot be directly observed (molecular-level particle motion), reflect on the outcomes of their manipulations, form conceptual understanding of particle behavior, and experimentally test these understandings in subsequent challenges. This cycle repeats across levels, deepening conceptual sophistication (Kolb, 1984).

### **Tier 3: Operational Frameworks**

Outcome-Based Teaching and Learning (OBTL) principles ensure alignment between pedagogical theory and curricular competencies (Sison et al., 2018). The game's learning outcome mechanics (LOMs) are explicitly designed to target specific competencies from the MATATAG Grade 6 Science curriculum, ensuring that gameplay directly supports achievement of standards-based objectives rather than generic skill development.

Bloom's Taxonomy guides cognitive level alignment (Anderson & Krathwohl, 2001). Lower-order objectives (remember, understand) are addressed through labeled particle representations and discovery mechanics; higher-order objectives (apply, analyze) are addressed through complex multi-phase challenges requiring strategic particle manipulation to solve environmental puzzles. This hierarchical alignment ensures the game scaffolds learning from foundational understanding to sophisticated application.

Iterative Design Principles and User-Centered Development Frameworks justify the two-phase approach (Section 2.5). Rather than conducting summative evaluation immediately after development, the methodology incorporates repeated cycles of playtesting, feedback integration, and refinement (Gee, 2003; Prensky, 2001). This approach recognizes that pedagogical theory informs initial design, but actual player experiences reveal implementation gaps; iterative cycles resolve these gaps before formal evaluation, ensuring the final game genuinely embodies the theoretical principles that justify its use (Sison et al., 2018).

In synthesis, these frameworks operate hierarchically: Primary theories explain the deep pedagogical mechanisms (why games can facilitate conceptual change through constructivist

interaction and multimedia learning); supporting theories specify motivational and experiential mechanisms (how games sustain engagement through challenge and experiential cycles); and operational frameworks translate these mechanisms into specific game features (which competencies to target and how to align gameplay to those targets). Iterative development ensures theoretical alignment, while structured evaluation validates achievement of alignment.

## 2.5 Conceptual Framework

The study adopts an expanded Input–Process–Output (IPO) model integrating both pedagogical foundations and iterative development practices. This framework visualizes how theoretical inputs are operationalized through development and evaluation processes to produce measurable outputs supporting student learning.

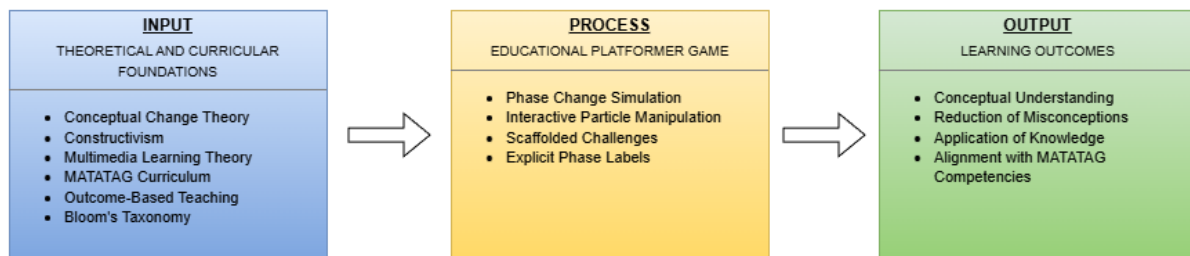


Figure 2.1: Conceptual Framework of the Study

### 2.5.1 Input Component

The Input component comprises three integrated elements:

1. **Theoretical Foundations:** Conceptual Change, Constructivism, Multimedia Learning, Game-Based Learning, and Experiential Learning theories (described in Section 2.4) provide the pedagogical justification for game-based intervention. These theories collectively explain why interactive gameplay can support understanding of abstract phase change phenomena when traditional instruction alone proves insufficient.
2. **Curricular Alignment:** The MATATAG Grade 6 Science competency on changes of state defines the content scope. Bloom's Taxonomy and Outcome-Based Teaching principles establish cognitive level expectations and learning outcome targets. This ensures the game reinforces competencies students are expected to master rather than tangential content.

3. **User-Centered Design Principles:** Iterative design, playtesting feedback loops, and engagement evaluation models (such as MEEGA+KIDS) guide the development process. These principles ensure the game is not only pedagogically sound in theory but also usable, engaging, and motivating in practice.

### 2.5.2 Process Component

The Process component is deliberately divided into two sequential phases, each with distinct purposes:

#### Iterative Development Phase (Internal Refinement)

- **Campaign testing:** Systematic playthrough of game levels to identify progression issues, accessibility barriers, and gameplay friction
- **Regression testing:** Verification that modifications made in response to feedback do not reintroduce previously resolved issues
- **Playtesting feedback integration:** Incorporation of player observations, difficulty reports, and suggestions into game mechanics and usability improvements
- **Gameplay refinement:** Iterative modification of particle mechanics, level difficulty curves, narrative elements, and UI design based on accumulated feedback

This phase prioritizes alignment between pedagogical theory (Inputs) and actual player experience. For example, if Constructivist principles suggest players should discover phase relationships through experimentation, but playtesting reveals players are confused by ambiguous feedback mechanisms, refinements clarify feedback without resorting to explicit instruction, preserving the constructivist approach while improving comprehension.

#### Final Evaluation Phase (Summative Assessment)

- **Structured play session using finalized build:** Controlled gameplay environment with Grade 6 participants
- **MEEGA+KIDS usability and engagement assessment:** Quantitative measurement of focused attention, satisfaction, perceived learning, and intrinsic motivation dimensions

- **Qualitative feedback collection:** Post-play interviews or surveys capturing student perceptions of learning value, difficulty appropriateness, and enjoyment

This phase validates whether the refined game achieves its intended educational and motivational objectives.

### **2.5.3 Output Component**

The Output component captures both developmental and evaluative results:

#### **Developmental Outputs**

- Improved gameplay stability and reduced technical issues
- Refined user experience with reduced navigation friction and clarity-enhanced feedback mechanisms
- Optimized difficulty progression supporting both accessibility and appropriate challenge

#### **Evaluative Outputs**

- Measured engagement levels across MEEGA+KIDS dimensions indicating learner motivation and perceived learning
- Evaluated usability confirming Grade 6 students can successfully operate game mechanics without excessive frustration
- Evidence of conceptual understanding gains through pre-post assessment comparisons (evaluated in later research phases beyond this thesis scope)

The framework illustrates a systematic progression: theoretical foundations inform initial design, iterative development resolves implementation gaps, and final evaluation confirms both the game's usability and the validity of the theoretical approach. Importantly, the separation of iterative development from summative evaluation ensures that classroom-ready game builds undergo rigorous user testing before formal assessment, maximizing likelihood that measured outcomes reflect genuine intervention effectiveness rather than usability limitations masking pedagogical potential.



## **CHAPTER 3**

### **RESEARCH METHODOLOGY**

#### **3.1 Research Design**

This study employed a developmental experimental research design integrating iterative game development practices with structured usability and engagement evaluation. The research was conducted in two major phases: (1) Iterative Development and Testing, and (2) Final Structured Evaluation.

The first phase focused on refining the educational platformer through multiple campaign and regression testing cycles. The second phase involved evaluating the finalized build using standardized usability and engagement instruments.

This design was selected to align with user-centered game development methodologies while maintaining academic rigor in assessing the educational game's usability and engagement potential.

#### **3.2 Development Framework**

The development of the educational platformer followed an iterative design approach grounded in user-centered development principles.

##### **3.2.1 Iterative Development Cycles**

The game underwent multiple structured development cycles consisting of:

1. **Build Implementation** – Addition or refinement of gameplay mechanics.
2. **Campaign Testing** – Structured playthrough sessions to identify usability issues, progression bottlenecks, gameplay friction, and feature clarity.
3. **Regression Testing** – Verification that newly implemented changes did not reintroduce previously resolved issues.
4. **Feedback Integration** – Refinement of mechanics, level design, UI clarity, and performance stability.

Each iteration was documented using testing logs detailing:

- Identified bugs
- Gameplay balancing concerns
- Player confusion points
- Feature stability issues
- Suggested improvements

Data from this phase were analyzed descriptively to track improvement trends across builds.

### **3.3 Participants**

Participants were selected using convenience sampling. Approximately 20–30 individuals were recruited through personal networks, academic peers, and referrals.

Participants were not restricted to Grade 6 students due to timeline and accessibility constraints. However, respondents were considered representative users capable of evaluating usability, engagement, and clarity of the educational mechanics.

Where participants were minors, parental consent was secured prior to participation.

Different participants may have participated across iterative testing cycles. However, the final structured evaluation phase was conducted using a stable release candidate build.

### **3.4 Research Instruments**

The study utilized both developmental testing tools and structured evaluation instruments.

#### **3.4.1 Campaign and Regression Testing Logs**

Structured testing logs were used to document:

- Bug occurrence frequency

- Severity classification (minor, moderate, critical)
- Gameplay progression issues
- UI clarity concerns
- Player navigation difficulties
- Performance stability

These logs enabled systematic tracking of improvements across development cycles.

### **3.4.2 MEEGA+KIDS Usability and Engagement Instrument**

The finalized version of the game was evaluated using the MEEGA+KIDS model. This instrument measures:

- Fun
- Focused attention
- Perceived learning
- Usability
- Challenge
- Satisfaction

Participants responded using a Likert-scale format.

### **3.4.3 Qualitative Feedback Questionnaire**

Participants provided open-ended responses regarding:

- Strengths of the game
- Areas requiring improvement
- Suggestions for refinement
- Perceived clarity of phase change concepts

### **3.5 Data Collection Procedure**

The research was conducted in two phases:

#### **3.5.1 Phase 1: Iterative Development Testing**

1. A development build was deployed.
2. Participants conducted structured play sessions.
3. Campaign testing logs documented usability and gameplay issues.
4. Regression testing verified stability after modifications.
5. Revisions were implemented before the next cycle.

This phase continued until a stable release candidate was achieved.

#### **3.5.2 Phase 2: Final Structured Evaluation**

1. Participants were briefed on the study purpose.
2. Consent was obtained.
3. Participants were given 20–30 minutes to interact with the finalized build.
4. Upon completion, participants completed:
  - MEEGA+KIDS instrument
  - Qualitative feedback questionnaire
5. Responses were collected via Google Forms.

### **3.6 Data Analysis**

Quantitative data from the MEEGA+KIDS instrument were tabulated and analyzed using Google Sheets.

The following descriptive statistics were computed:

- Mean
- Standard deviation
- Frequency distribution
- Percentage scores

Results were interpreted to determine overall engagement and usability levels of the finalized build.

Data from campaign and regression testing logs were analyzed descriptively to identify:

- Reduction in bug frequency across iterations
- Improvement in gameplay stability
- Decrease in recurring usability issues

Qualitative responses were grouped into thematic categories to identify common patterns in player feedback.

No inferential statistical comparisons between builds were conducted, as iterative development phases involved varying participant groups.

### **3.7 Ethical Considerations**

Participation in the study was voluntary. Participants were informed of the study's purpose and their right to withdraw at any time without penalty.

No personally identifiable information was included in the analysis. For participants under the age of 18, parental consent was secured prior to participation.

All collected data were used solely for academic research purposes.

**CHAPTER 4**

**DESIGN AND IMPLEMENTATION**

## CHAPTER 5

### GAME SYSTEM

#### 5.1 Software Overview

*Wiz's Journey: Mastering the States of Matter* is developed using Unity 2022 LTS, a cross-platform game engine selected for its extensive capabilities in 2D game development, robust documentation, and the development team's prior expertise. The game is designed as a 2D platformer that integrates educational content with interactive gameplay mechanics, allowing students to progress through levels while encountering learning outcomes related to phase changes of matter.

The game is deployed across multiple platforms to ensure widespread accessibility and reach diverse student populations. The primary deployment target is Android mobile devices, distributed through Itch.io, a popular game distribution platform that allows easy access without requiring app store approval processes. A secondary web-based version is deployed via Unity Play, allowing students to play directly in web browsers on computers and tablets without installation requirements. This multi-platform approach reflects the varying technological infrastructure available to schools and students in the Philippines and aligns with the goal of making the educational game accessible regardless of device availability.

The development timeline began in December 2025 at the conclusion of the first trimester and extended through March 2026, spanning the academic term break and into the second trimester. This timeline was established to accommodate the academic calendar constraints and allowed the development team to complete the game while managing other coursework obligations. By the time development concluded in March 2026, the target Grade 6 students had already completed their first quarter instruction on phase changes under the MATATAG curriculum, positioning the game as a supplementary learning reinforcement tool rather than a primary instructional resource.

The game architecture incorporates both lower-order and higher-order mechanics aligned with the selected learning outcomes. Lower-order mechanics focus on foundational understanding of the four phase changes—melting, freezing, evaporation, and condensation—using visual representations and simple interactions. Higher-order mechanics build upon this foundation to address more complex concepts such as the relationship between heat transfer and state changes, the particulate nature of matter during phase transitions, and critical thinking about real-world applications of phase changes. The system integrates a particle lattice visualization system to support students' conceptualization of microscopic-level processes underlying

macroscopic observations, addressing a common misconception identified in science education research.



**CHAPTER 6**  
**RECOMMENDATIONS**